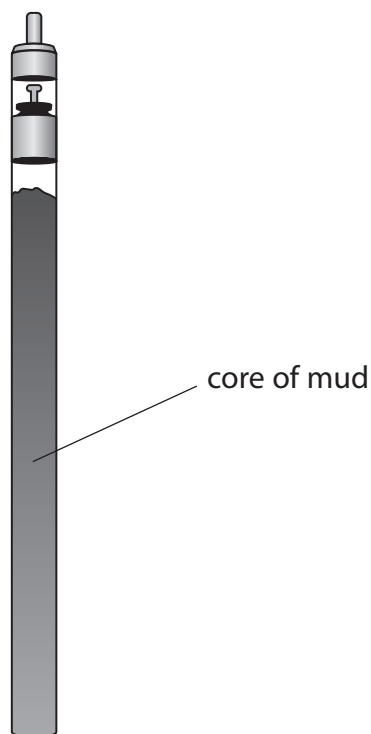


- 4 Scientists have used specialised apparatus to remove cores of mud from a tropical rainforest in the Republic of the Congo.

The diagram shows the apparatus containing a core of mud.



- (a) The core of mud removed has a diameter of 80 mm and a length of 900 mm.

Which is the volume of this core of mud?

Use the formula: $V = \pi r^2 l$

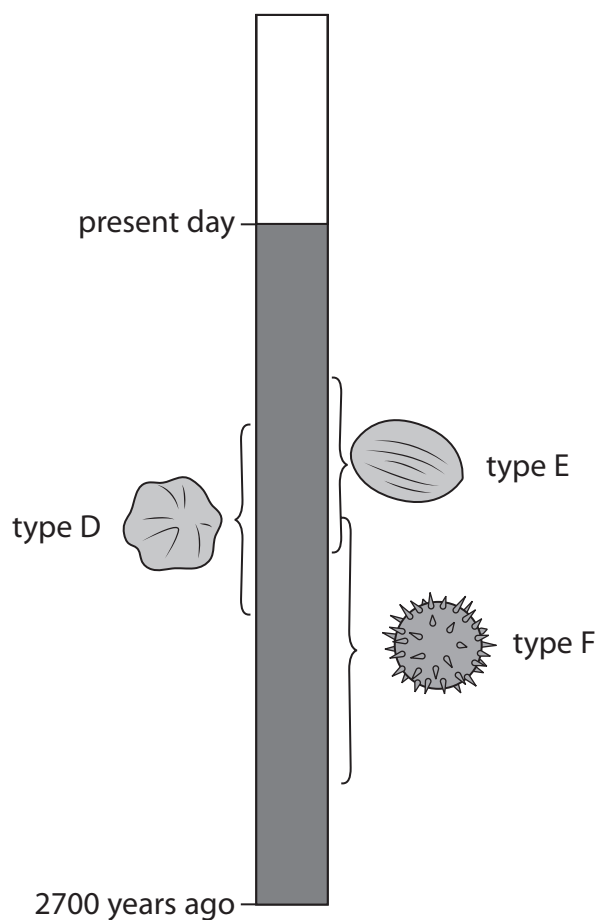
Use $\pi = 3.142$

(1)

- ☐ **A** 4524 cm³
- ☐ **B** 4525 cm³
- ☐ **C** 18 097 cm³
- ☐ **D** 18 098 cm³

- (b) The scale diagram shows where three types of pollen grain were found in a core of mud.

The depth of the mud is proportional to how long ago the mud was deposited.



- (i) Calculate how many years the plants producing type D pollen grains were present in this rainforest.

(2)

Answer years

- (ii) Explain the distribution of these three types of pollen grain in this mud column.

Use the information in this diagram to support your answer.

(4)

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(Total for Question 4 = 7 marks)